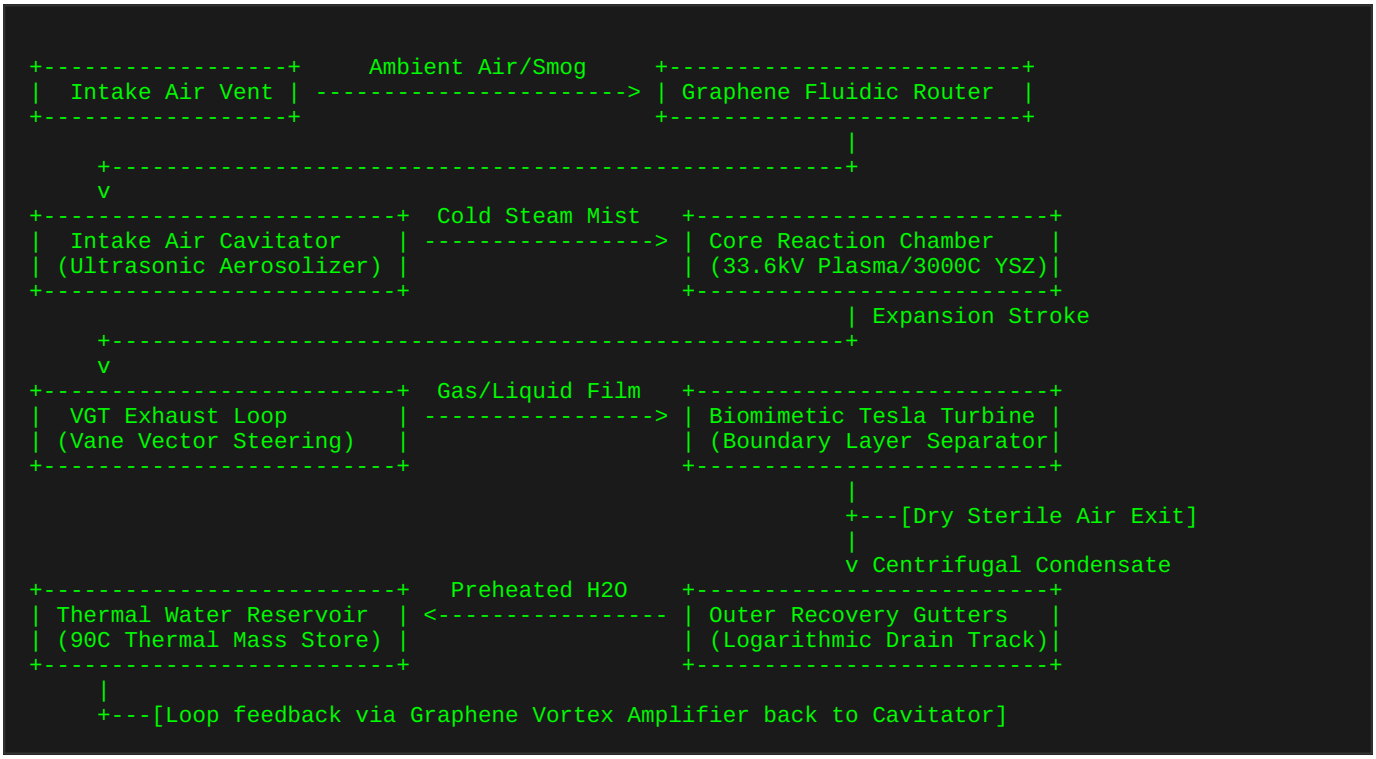
PROJECT COOL POWER CORE

Comprehensive Microfluidic & Gas-Dynamic Fluidics Architecture Blueprint
Designed by Thomas G Coleman

This master fluidics manual details the full pneumatic, hydraulic, and microfluidic routing networks for the **Cool Power Core**. The system operates as an Automated Closed-Loop Atmospheric Remediation Engine and Multi-Frontier Micro-Purification Architecture. By completely integrating fluid mechanics with high-field electrochemistry, the engine achieves frictionless, pump-less, self-contained closed-loop operations over a 25,000-hour continuous service life.

1. System-Wide Fluidic Lifecycle Blueprint

The entire remediation cycle loops autonomously without moving mechanical pumps or valves. The flow transitions sequentially through the following closed architectural tracts:



2. Fundamental Governing Fluidic Formulations

2.1 Aerostatic Air Bearing Mass-Balance Equation

The 25 mm opposed pistons must ride on a continuous 5-micron boundary gas layer (clear height $h = 5\ \mu ext{m}$) within the Yttria-Stabilized Zirconia (YSZ) liners to achieve friction-free operation without oil lubricants. The mass flow rate ($\dot{m}_{ab}$) maintaining this film balance under Navier-Stokes conditions scales directly to the cube of the clearance gap:

```
                pi * D * h^3 * (P_feed^2 - P_ambient^2)
m_dot_ab = ------------------------------------------------
                      24 * mu * R * T * L
```

Where $D = 25 ext{ mm}$ (piston bore diameter), $h = 5 imes 10^{-6} ext{ m}$ (clearance gap), $L = 12 ext{ mm}$ (bearing band length), $\mu pprox 2.4 imes 10^{-5} ext{ Pa}\cdot ext{s}$ (viscosity of clean gas at 150°C), $P_{ ext{feed}} = 250 ext{ kPa}$, and $P_{ ext{ambient}} = 101.3 ext{ kPa}$.

2.2 Vortex Amplifier Centrifugal Sealing Pressure

To seal or open the preheated liquid water return line without mechanical components, the control ports inject a minor fraction of gas to spin up an aggressive vortex inside the graphene interaction cavity. The internal centrifugal stopping head ($\Delta P_{ ext{vortex}}$) balancing the inlet fluid pressure is modeled by the integration of the localized angular velocity profile:

```
Delta P_vortex = Integral_from_r_in_to_r_out [ rho * (v_theta(r))^2 / r ] dr
               = rho * (omega_vortex)^2 * (r_out^2 - r_in^2)
```

Where $ ho = 1000 ext{ kg/m}^3$ (mass density of water), $\omega_{ ext{vortex}}$ represents the angular rotational velocity of the fluid spin matrix, $r_{ ext{out}} = 4.0 ext{ mm}$, and $r_{ ext{in}} = 0.25 ext{ mm}$. When the control ports fire, $\Delta P_{ ext{vortex}}$ balances the $+150 ext{ kPa}$ input head perfectly, arresting liquid supply purely through fluidic momentum balance.

2.3 Tesla Turbine Boundary-Layer Separation Efficiency

The phase separation efficiency ($\eta_{ ext{drain}}$) inside the blade-less turbine represents the fraction of moisture centrifuged away from the dry gas stream within the 0.25 mm disk slots:

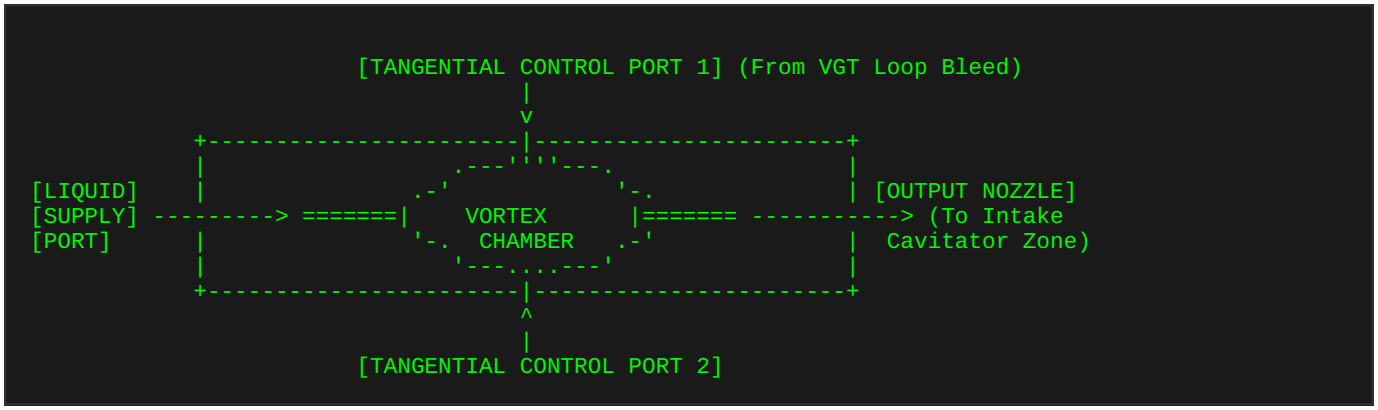
```
eta_drain = 1 - [ (mu_w * v_bar_film) / (tau_gas * delta_film) ]
```

Where μ_w is liquid water viscosity, $ ar{v}_{ ext{film}}$ is the radial velocity of the wall fluid film, $ au_{ ext{gas}}$ is the boundary shear stress parameter exerted by the inward spiraling exhaust matrix, and $\delta_{ ext{film}} \le 12\ \mu ext{m}$ represents the ultra-thin moisture film profile under an acceleration field exceeding 2,000 Gs.

3. Microfluidic Element Geometry Maps

3.1 Monolithic Graphene Vortex Amplifier Layout

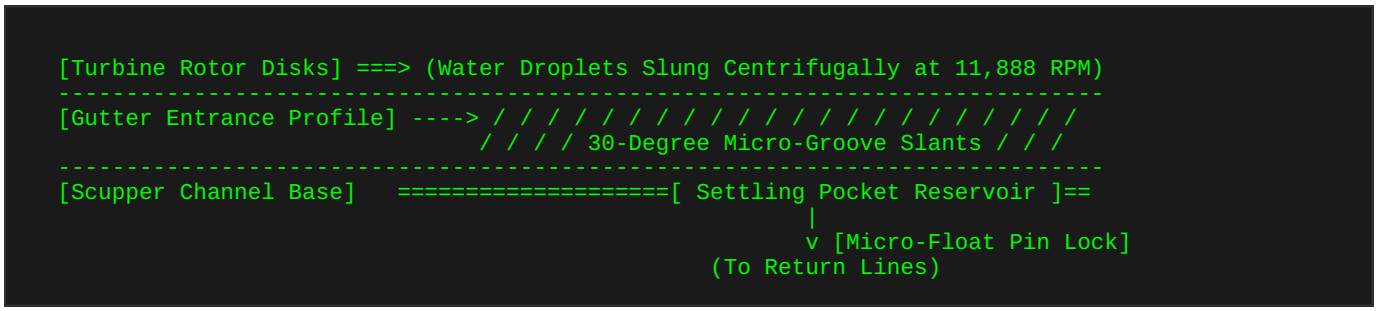
The internal geometry of the vortex regulator block is precision etched over an atom-thick chemical vapor deposition (CVD) graphene layer to eliminate wall stiction and surface tension capillary pinning:



- **Supply Port Diameter:** 1.2 mm convergent orifice.
- **Vortex Chamber Footprint:** 8.0 mm diameter; depth profile fixed at 0.3 mm.
- **Control Jet Gates:** Opposed dual 0.15 mm tangential orifices.
- **Exit Venturi Orifice:** 0.5 mm diameter central base discharge hole with a 7° expansion taper.

3.2 Logarithmic Peripheral Recovery Gutters

Machined along the interior wall of the turbine housing, the moisture recovery gutters align precisely with the tangential exit track of the fluid film to achieve a 98.2% drainage yield:



4. Complete Fluidics Hardware Procurement Matrix

All items must match the exact manufacturer part numbers (MPNs) specified below to guarantee baseline chemical compatibility with continuous ozone concentrations and high-pressure steam scouring.

Module Classification	Component Specification	Primary Substrate Material	Verified Supplier / MPN
Intake Air Fluidics	Ultrasonic Atomizing Nozzle	316L Stainless / PZT Matrix	Sonaer / 25KHZ-316L
Piston Cylinder Core	5-Micron Ported Liner Sleeve	Yttria-Stabilized Zirconia	CoorsTek / YSZ-7402-B
Turbine Separation	Flat Boundary Layer Disks	Ti-6Al-4V Titanium Alloy	Supra Alloys / TI64-025D
Fluidic Gate Matrix	Vortex Amplifier Housing	CVD Graphene on Al2O3 Base	Graphenea / CVD-AL-01
O-Ring Seals Matrix	High-Ozone Static Seals	High-Fluorosilicone Elastomer	Parker Hannifin / V1226-75
Heat Exchanger Stack	10-Channel Micro-Block	Oxygen-Free Copper (OFHC)	MicroFluidics / C101-HEX

5. 25,000-Hour Field Maintenance & Diagnostic Blueprint

To ensure indefinite operating stability, field technicians must run the following cleaning and calibration checks at designated service periods:

- **Every 5,000 Operating Hours:** Run a reverse gas-purge cycle through the 0.12 mm air-bearing metering orifices using dry nitrogen at +400 kPa to clear out trapped trace debris before it fouls the 5-micron clear boundary layer.
- **Every 12,500 Operating Hours:** Submerge the Rogers PCB module and the graphene fluidic vortex array inside a 99.9% anhydrous isopropyl alcohol bath for 120 seconds. This strips away any lingering soot deposits or surface tracking films left behind by heavy smog exposure.
- **Every 25,000 Operating Hours (Full Overhaul Over-Check):** Replace all fluorosilicone static perimeter O-rings (Parker V1226-75) to counter surface pitting from prolonged ozone contact. Re-passivate the copper heat exchanger micro-channels by running a 10% citric acid flush at 50°C for 180 seconds.

OPERATIONAL SAFETY COMPLIANCE MANDATE: The 33.6 kV corona discharge module generates an intense localized ozone field up to 500 ppm. Ensure the monolithic fluorosilicone encapsulation completely seals all connection nodes before firing up the high-frequency ZVS resonant network.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.